

Lab 3: Architecture Justification

Problem Statement #11: Telemedicine Slot Booking & Prescription Portal

Architecture Selection

We chose **Layered Architecture** for the **Telemedicine Slot Booking & Prescription Portal**.

Style	Fit for the Telemedicine Portal
Layered (chosen)	UI, logic and data map naturally to the portal. Simple to build, test and secure.
Microservices	Good scaling and fault isolation, but too much complexity (many services, network calls, data consistency) for 5 core features.
Client-Server	Simple, but one central server with mixed logic is a single point of failure and has weaker separation of sensitive data.

Two Reasons for the Choice

- **Natural fit with the workflow.** A patient or doctor uses a screen (Presentation), which triggers booking, consultation or prescription logic (Business), which reads or saves records (Data). This maps one-to-one onto the three layers.
- **Small, well-defined scope.** The portal only needs five core features (view specialties, book slots, get a room link, sign prescriptions, download PDFs). Layers keep each component easy to build, test and replace without the overhead of microservices.

Security Advantage

Only the Business layer can reach the Database, so the UI never touches patient data directly. Login and doctor-license checks (Authentication Service) and signing (Prescription Service) are enforced in one place, and encryption (TLS 1.3 in transit, AES-256 at rest) is applied at the layer boundaries. This gives a small, controlled attack surface for sensitive health data.

Performance Benefit

Doctor specialties and slot lists are read often but change rarely, so the Business layer can cache them and avoid a database query on every page load. The Consultation Service only generates the encrypted room link, so heavy video traffic does not pass through the portal layers.

Components and Interfaces (see diagram)

Components: Portal UI, Appointment Manager, Consultation Service, Authentication Service, Prescription Service, Database.

Interface	Provided by	Required by	Data flow
IAuthentication	Authentication Service	Portal UI	Login credentials, session token
ILicenseValidation	Authentication Service	Prescription Service	Doctor license number, valid / invalid
IBooking	Appointment Manager	Portal UI	Specialty search, slot list, booking request
IConsultation	Consultation Service	Appointment Manager	Booking ID in, encrypted room link out
IPrescription	Prescription Service	Portal UI	Prescription details, signed PDF
ISlotStore	Database	Appointment Manager	Slots, bookings, room links
IUserStore	Database	Authentication Service	Patient and doctor records
IRecordStore	Database	Prescription Service	Prescriptions, clinical notes